



Calculating with Units Tutorial

1. Units multiply and divide like numbers, but don't combine with the numbers.
2. To combine and cancel units, convert them.
3. Use equalities to convert units.
4. Linear, squared, and cubed units DO NOT have the same conversions.
5. Keep track of all units.

Commentary:

1. A simple example is $(2 \text{ mm})(2 \text{ mm}) = 4 \text{ mm}^2$. The 2's multiply and the mm's multiply, but there's no way to combine a 2 and a mm into some more compact form. This multiplication with units is identical to doing the algebraic calculation $(2x)(2x) = 4x^2$. The 2's multiply and the x's multiply, but they don't combine.

2. If you want to multiply 2 mm and 0.2 cm, you could write $(2 \text{ mm})(0.2 \text{ cm}) = 0.4 \text{ mm} \cdot \text{cm}$ [analogous to $(2x)(0.2y) = 0.4xy$]. This is perfectly correct, but the mm·cm unit of area, literally the area of a rectangle that is 1 mm by 1 cm, is not one that is commonly used. In general, it is preferable to report answers in more compact units, like mm^2 or cm^2 . To do so, one can either convert the 2 mm into centimeters or the 0.2 cm into millimeters, i.e. $(0.2 \text{ cm})(0.2 \text{ cm}) = 0.04 \text{ cm}^2$ or $(2 \text{ mm})(2 \text{ mm}) = 4 \text{ mm}^2$. These equivalent calculations are both correct and conform to convention.

3. In converting a quantity like 2 mm to some other unit of length, you must not alter this quantity. It represents a physical length, and after changing the number and units you should have an expression that represents the same physical length. The only quantity with which you can multiply 2 mm with no change is 1. Of course, the calculation $2 \text{ mm} \cdot 1 = 2 \text{ mm}$ isn't especially helpful. This is where equalities come in. To convert mm's to cm's we need some, indeed any, equality relating the units: $10 \text{ mm} = 1 \text{ cm}$, $370 \text{ mm} = 37 \text{ cm}$, $6 \text{ mm} = 0.6 \text{ cm}$, etc. We can then use this equality to create a new expression that is equal to 1 and will be useful in unit conversion. For example, take the equality $10 \text{ mm} = 1 \text{ cm}$, and divide both sides by 10 mm:

$$\frac{10 \text{ mm}}{10 \text{ mm}} = \frac{1 \text{ cm}}{10 \text{ mm}} \rightarrow 1 = \frac{1 \text{ cm}}{10 \text{ mm}}$$

Given that 2 mm is unchanged if we multiply it by $1 = (1 \text{ cm}/10 \text{ mm})$:

$$2 \text{ mm} = 2 \text{ mm} \cdot 1 = 2 \text{ mm} \cdot \frac{1 \text{ cm}}{10 \text{ mm}} = \frac{2 \text{ cm}}{10} = 0.2 \text{ cm}$$

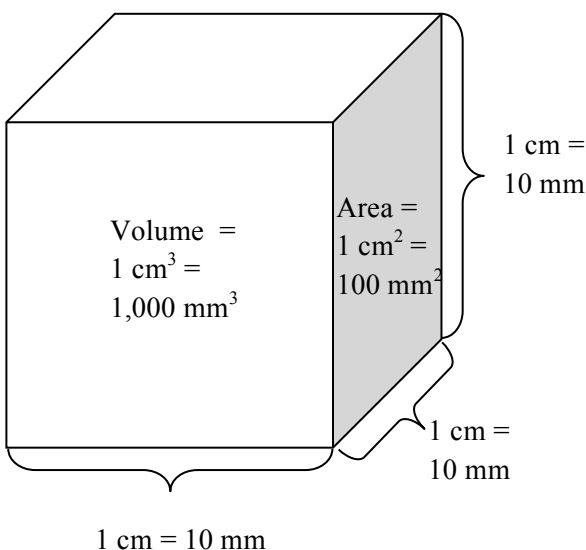
This is the conversion we wanted. What if, instead of multiplying 2 mm by $1 = (1 \text{ cm}/10 \text{ mm})$, we had multiplied it by $1 = (10 \text{ mm}/1 \text{ cm})$? We then would have gotten:

$$2 \text{ mm} = 2 \text{ mm} \cdot 1 = 2 \text{ mm} \cdot \frac{10 \text{ mm}}{1 \text{ cm}} = \frac{20 \text{ mm}^2}{1 \text{ cm}} = 20 \frac{\text{mm}^2}{\text{cm}}$$

This is perfectly correct, but the units of length, mm^2/cm , are again unconventional, and not useful here, because the goal was to convert mm to cm. We have made no mathematical mistakes, but have not converted 2 mm into the units we wanted. This leads us to a general rule of thumb: the units we are converting into should be in the numerator of the expression for 1 and the units we are converting out of should be in the denominator. When converting from mm's to cm's we should multiply by

$1 = (1 \text{ cm}/10 \text{ mm})$, when converting from cm's to mm's we should multiply by $1 = (10 \text{ mm}/1 \text{ cm})$.

4. Though cm , cm^2 , and cm^3 are all related, they are different units, and they measure different physical quantities: length, area and volume, respectively. Their conversions to mm, mm^2 , and mm^3 are also related but different. The cube below illustrates the differences:



The sides of the cube are each $1 \text{ cm} = 10 \text{ mm}$, the area of each face is $(1 \text{ cm})^2 = (10 \text{ mm})^2$ or $1 \text{ cm}^2 = 100 \text{ mm}^2$, and the volume of the cube is $(1 \text{ cm})^3 = (10 \text{ mm})^3$ or $1 \text{ cm}^3 = 1,000 \text{ mm}^3$. Though $1 \text{ cm} = 10 \text{ mm}$,

$1 \text{ cm}^2 \neq 10 \text{ mm}^2$ and $1 \text{ cm}^3 \neq 10 \text{ mm}^3$. The correct conversions for squared and cubed units can always be found from the linear units by taking the linear equality and squaring or cubing both sides. For example, $1 \text{ mile} = 1.6 \text{ kilometers}$. So $(1 \text{ mile})^2 = (1.6 \text{ kilometers})^2$ or $1 \text{ square mile} = 2.56 \text{ square kilometers}$. Failing to use the proper conversion between squared and cubed units is a common mistake. Proceed carefully when doing so.

5. Keeping track of units at all stages of a calculation serves two purposes. First, explicitly writing out the units when you start a calculation will help to confirm that you've put the correct quantities into your calculation in the correct places. For example, suppose that you want to know the distance light travels in 5 minutes. Trying to multiply the speed of light by this time by writing $(3 \times 10^8 \text{ m/sec})(5 \text{ minutes})$ will show you that you can't just multiply 3×10^8 times 5 to get the answer, but should convert the 5 minutes to 300 seconds to get an answer in meters. Second, since you are multiplying and dividing the units alongside the multiplication and division of the numbers, the units give you a way of detecting errors – if you end up with some unexpected units at the end it means you may have made some mistake in your calculation. In this example, if you accidentally divide the speed by the time instead of multiplying, you will get $(3 \times 10^8 \text{ m/sec})/(300 \text{ sec}) = 10^6 \text{ m/sec}^2$, which you can tell from the units is not a distance.